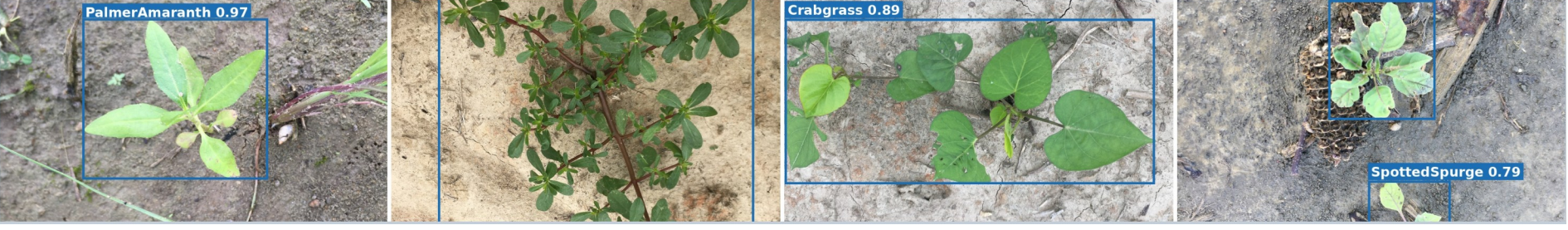


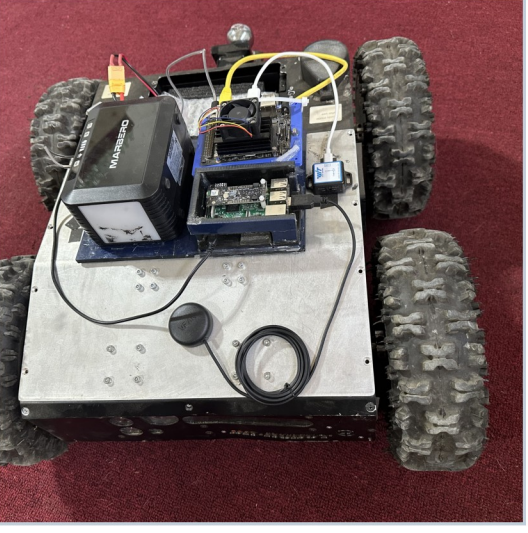
On one platform, agents collect their own data, review one another's work and train without an operator. The system ran for fifteen unattended rounds; this sheet reports how the agents coordinate and what the collection produced.

Weed Detection



The corpus exceeds 150,000 labelled images. Two agents operate on it unprompted: one harvests and audits imagery, the other trains a detector small enough for the robots.

Rover



A four-wheel skid-steer the width of a crop row carries a Jetson, a Raspberry Pi and its own battery. Camera, GPS and two IMUs stream live over one API key.

Laser Weed Control Robot



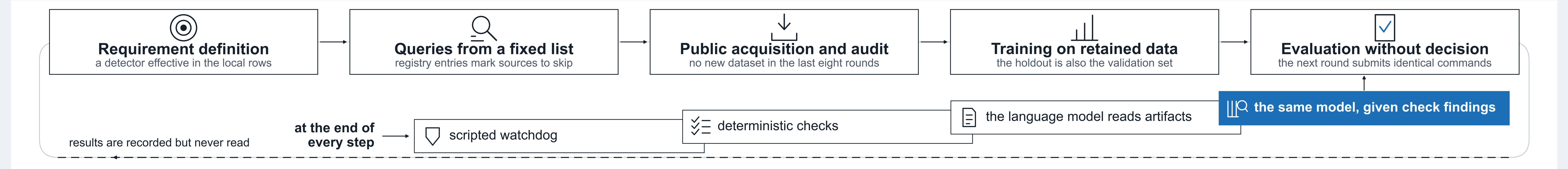
A YOLO detector localises each weed; two galvanometer lasers aim and fire, destroying it in place. No chemical is applied and nothing contacts the adjacent crop.

Humanoid and Quadruped



This project pairs a humanoid with a quadruped to record terrain that wheels do not reach. It adopts the Rover's ingest contract, so every session the robots open is governed.

Figure 1. Every project on the platform is created by the same procedure and registers its data under one ingest contract. The harvesting and training agents have so far run on one of them.



Agent Workflow Diagram

Figure 2. A requirement enters. The harvest issues fixed queries without gap analysis. Audited data train the model; the holdout is also the validation set and selects the reported epoch. Evaluation parses a metric; no decision follows. The reviewer runs after every step; the dashed edge is unimplemented.

ONE PLATFORM SUPPORTS ANY ROBOT OR DATASET

- **agentAI is one platform for every laboratory robot**
A robot, drone or sensor rig is a project with its own keys, members and access-controlled dataset registry. Agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.
- **Both the Rover and the laser robot transmit data**
The Rover is a four-wheel skid-steer streaming camera, GPS and two inertial measurement units. The Laser Weed Control Robot carries two galvanometer lasers on independent channels and two cameras, one over the work zone and one forward.
- **Each drive becomes a dataset at its start**
The platform registers a vehicle session as a governed dataset during the drive. Frames and sensor batches arrive over one API key. The laser robot transmits no position.
- **Every data type and domain forms its own project**
On upload, a project returns an automatic diagnosis of the data before the next step is chosen. Weed detection is one project; the Rover and the laser robot are two others.



■ **The platform is live; the QR code links to it**
lab-b660m-c.tailfa6424.ts.net
Authentication requires an institutional Google account.

OPEN-WEIGHT MODELS REVIEW IN TIERS

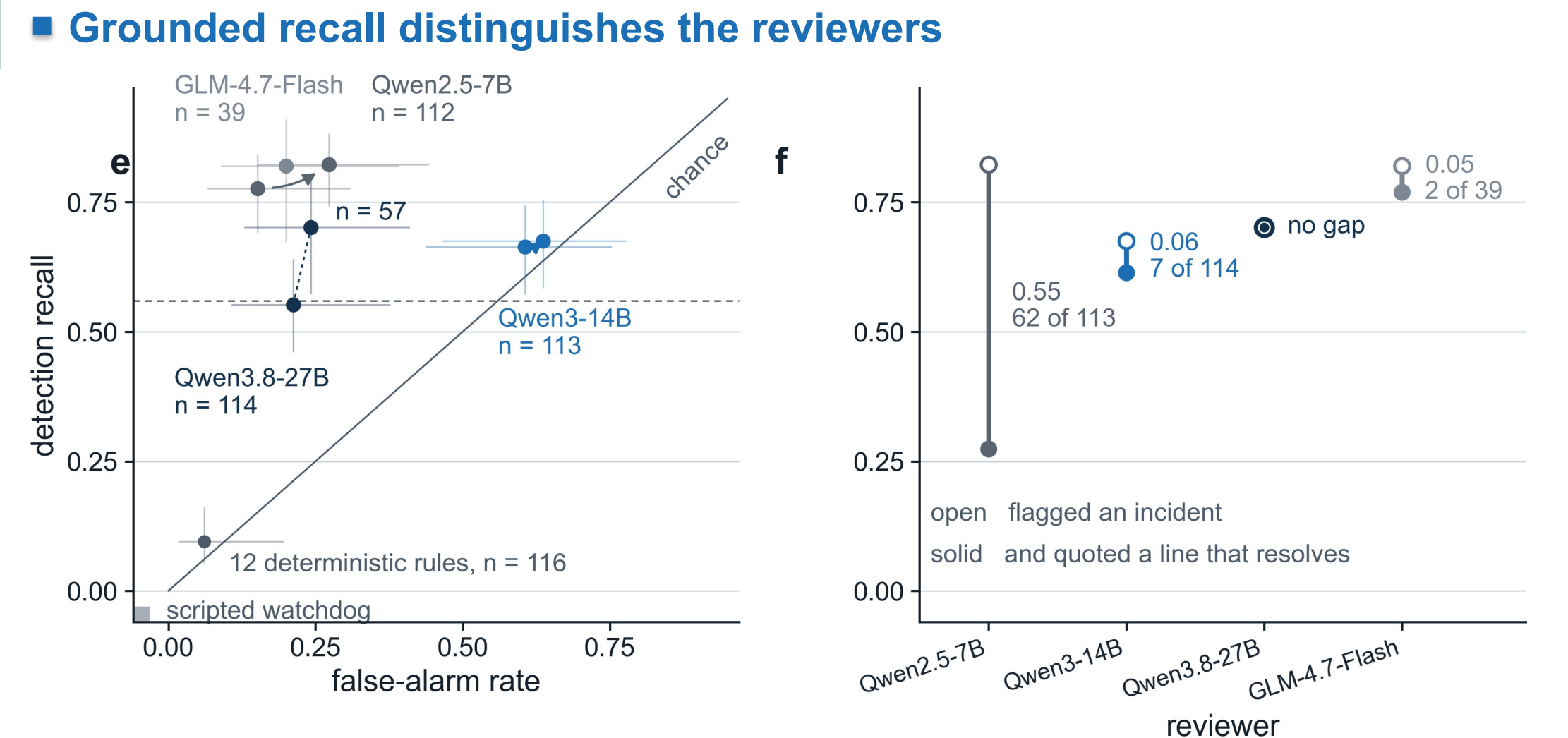


Figure 3. (e) Recall and (f) grounded recall on one frozen corpus of real incidents from this project's record, scored by its own scorer. Each tail is the language model on artifacts with the twelve checks' findings withheld; each head is the same model supplied with them. The dotted line has no arrowhead: its ends were scored on different cases, the 27 B head on 57 incidents and its tail on 114.

- Every reviewer is an open-weight model served from a self-hosted cluster. The higher tier adds grounding, not parameter count: supplying the deterministic tier's findings to the model yields +0.149 recall at 27 B and +0.012 at 14 B.
- **Tiered review is invoked at the end of every step**
The tiers are a scripted watchdog, twelve deterministic checks, a language model reading the artifacts with those checks withheld, then the same model supplied with them. On 149 frozen cases the scripted watchdog returned no decision at all.

A JUDGEMENT COUNTS ONLY WHERE IT RUNS

A judgement counts only where it was produced	
On the cluster	
plan the next step	qwen3.8 27B
review in depth	glm-4.7-flash
hard one-off reasoning	deepseek-v3 671B
decide what to collect	gemma 4
judge dataset quality	qwen2.5 7B
label assistance	minicpm-v
On the lab box	
turn intent into a project	qwen2.5 3B
summarise a dataset	qwen2.5 3B

Figure 4. The eight roles the router places, read from its own table. A judgement is authoritative only when it was produced on the cluster; the same judgement produced on the laboratory box returns flagged as a draft, so the placement rule travels with the answer rather than living in a configuration block. The flag exists because nine campaign reviews were once produced by a 4.7 GB model on the laboratory box while 458 GB of verified weights sat unused on the cluster, and nothing in the returned result said so.

- A router places each role on a model and a machine. Judgement produced on the cluster is returned authoritative; the same judgement produced on the laboratory box is returned marked a draft, so the rule travels with the answer.
- **A language model's falsifiable hypothesis was false**
Harvested boxes take their class from the source website. A language model proposed that as the cause, stated its falsifying condition first, and a human approved. Four runs falsified it. The cost was density: nine boxes per image against under two.

A JUDGEMENT COUNTS ONLY WHERE IT RUNS (CONT.)

- **Compounding was the premise of the unattended loop**
One agent collects and one trains, and each runs without an operator. Fifteen rounds ran that way. The training corpus did not change after round three, yet the ledger and every round report recorded each round as new work.

WHAT THE COLLECTION PRODUCED

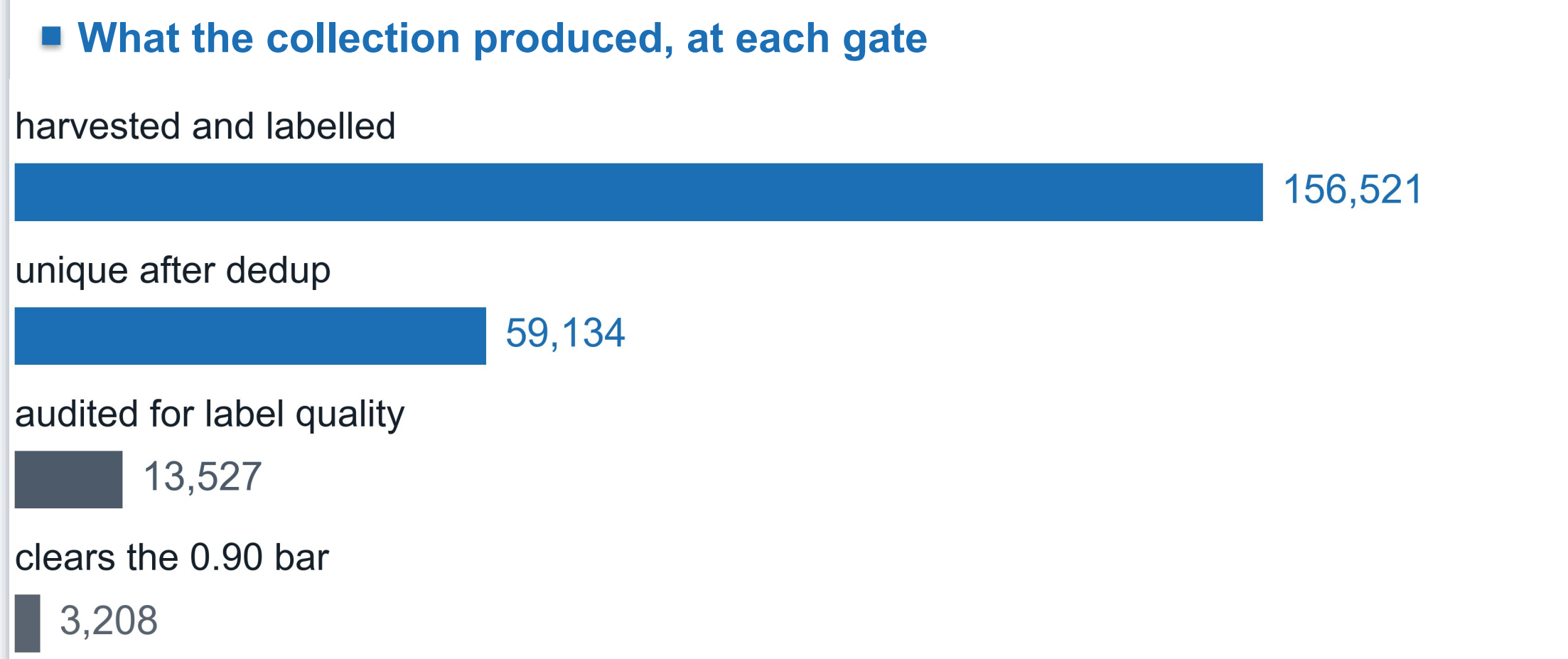


Figure 5. One of six audited harvested sources clears the label-precision bar. The audit probe reads 1.000 on human-labelled CottonWeedDet12, so the low scores are the data.

- Automatic labelling wrote 156,521 images. Cross-dataset deduplication leaves 59,134 distinct ones, and of the 13,527 audited for label precision only 3,208 clear the bar set for admission.
- **The training configuration was not the constraint**
Augmentation, replaying known species and completing labels were each tried and each left the same gap. What remains is the automatic labeller's boxes, and the audit above measures them.
 - **Geometry rejects boxes the language model accepts**
The rule discards boxes lying too far from the trusted data. Presented with a candidate box, Gemma 4 rejects none of the false boxes, so the criterion is DINOv2 geometry. Fifty-one datasets were removed.